# AERODYNAMICS PROJECT

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**Free Stream Velocity: 125 m/s**
**Chord Length: 1.25 m**
**Maximum Thickness Ratio: 9.5%**
**Maximum Camber Ratio: 3%**

# Flow Simulation over a Joukowski Airfoil

## Introduction

Airfoil aerodynamic performance plays a critical role in lift generation and aircraft efficiency. In this project, the potential flow over a **Joukowski airfoil** is simulated using analytical methods implemented in MATLAB. The Joukowski transformation allows mapping a circular cylinder in the complex plane into an airfoil shape, enabling closed-form solutions for velocity, pressure, lift, and moment coefficients.

The objective of this project is to analyze the aerodynamic characteristics of a Joukowski airfoil with **9.5% maximum thickness** and **3% maximum camber** under inviscid, incompressible flow conditions.

## Problem Definition and Given Data

The following parameters were specified:

- Free-stream velocity:

$$V_\infty = 125 \text{ m/s}$$

- Airfoil chord length:

$$c = 1.25 \text{ m}$$

- Maximum thickness to chord ratio:

$$t/c = 9.5\%$$

- Maximum camber to chord ratio:

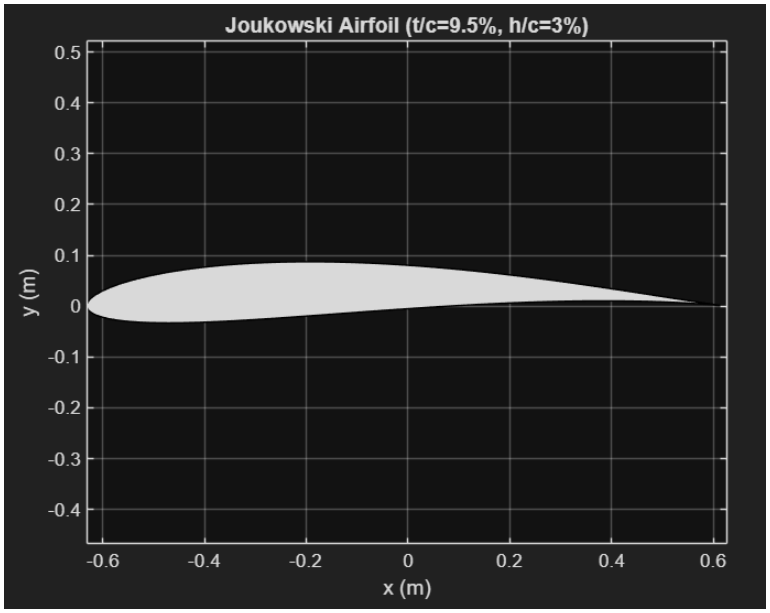
$$h/c = 3\%$$

- Angle of attack:
  - $\alpha = 6°$ for flow visualization
  - $\alpha = -5°$ to $10°$ for lift and moment analysis

## Airfoil Geometry Generation

The airfoil geometry was generated using the **Joukowski transformation**, which maps a shifted circular cylinder into an airfoil shape. The amount of thickness is controlled by the horizontal displacement of the circle, while camber is introduced by a vertical shift.

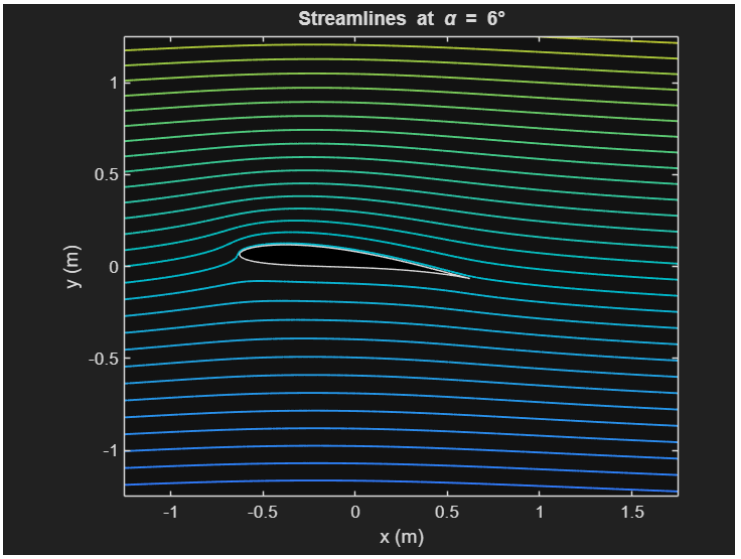
**Results:**




- The generated airfoil shows a **smooth leading edge** and a **sharp trailing edge**, which is characteristic of Joukowski airfoils.
- The **maximum thickness** occurs near the forward portion of the chord.
- The **positive camber** causes an upward curvature of the mean camber line, enabling lift generation at zero and small angles of attack.

**Streamline Pattern at $\alpha = 6^{\circ}$**

The streamlines were computed by solving the complex potential of uniform flow combined with circulation and applying the Joukowski transformation.




**Results:**

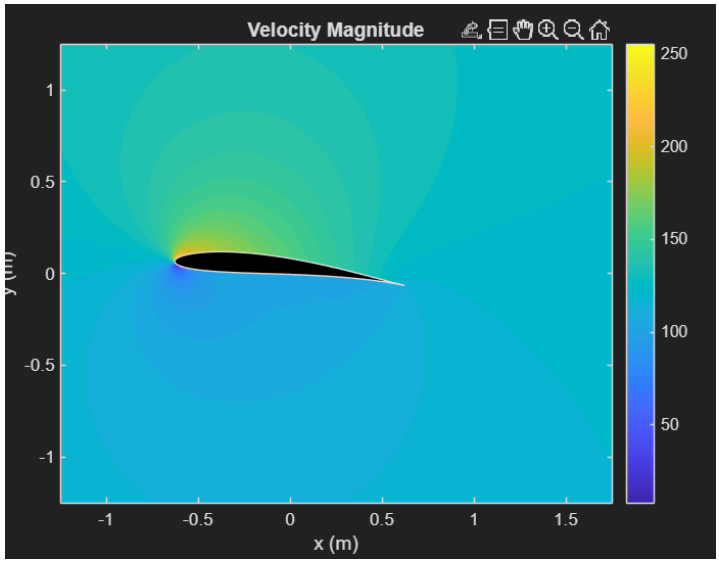
- Streamlines are **compressed above the airfoil** and **expanded below**, indicating lift generation.
- A stagnation point is visible near the lower leading edge.
- The streamline curvature increases near the leading edge due to strong acceleration of the flow.

This behavior is consistent with **classical potential flow theory**.

**Velocity Distribution at $\alpha = 6^{\circ}$**

The velocity magnitude was obtained from the derivative of the complex potential and transformed to the physical plane.

**Results:**



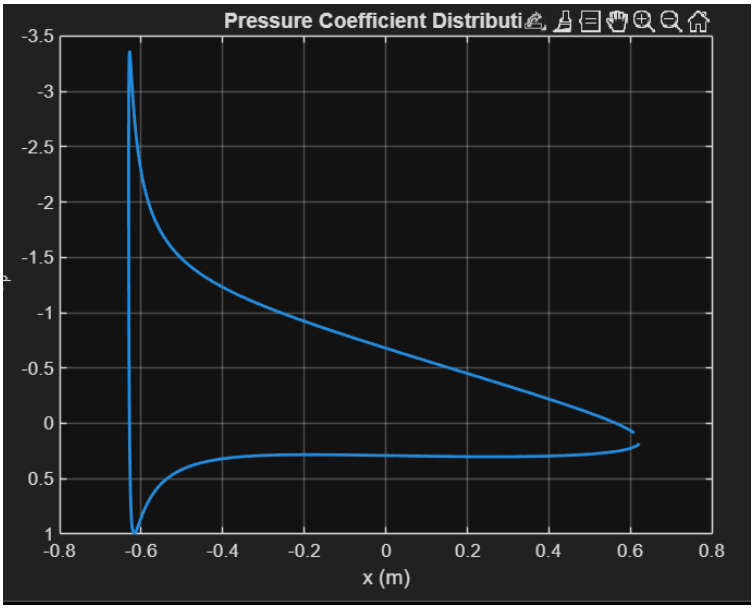

- The **maximum velocity occurs near the upper leading edge**, reaching values significantly higher than the free-stream velocity.
- Lower surface velocities remain comparatively lower.
- This velocity asymmetry explains the pressure difference between the upper and lower surfaces.

**Pressure Coefficient Distribution at $\alpha = 6^{\circ}$**

The pressure coefficient was computed using:

$$C_p = 1 - \left(\frac{V}{V_\infty}\right)^2$$

**Results:**



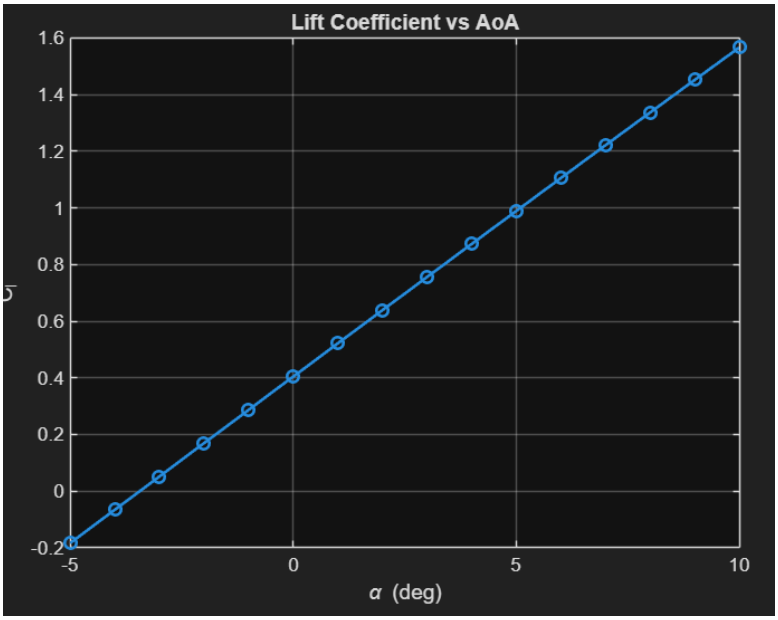

- A **strong suction peak** (highly negative $C_p$) appears near the upper leading edge.
- The pressure gradually recovers toward the trailing edge.
- The lower surface shows higher pressure values, contributing positively to lift.

This pressure distribution is physically realistic for a cambered airfoil at positive angle of attack.

**Lift Coefficient Variation with Angle of Attack**

The lift coefficient was calculated for angles of attack from $-5°$to $10°$.

**Results:**




- The lift coefficient increases **linearly** with angle of attack.
- A **positive lift at zero angle of attack** is observed due to camber.
- The slope of the $C_l$–$\alpha$ curve agrees with thin airfoil theory.

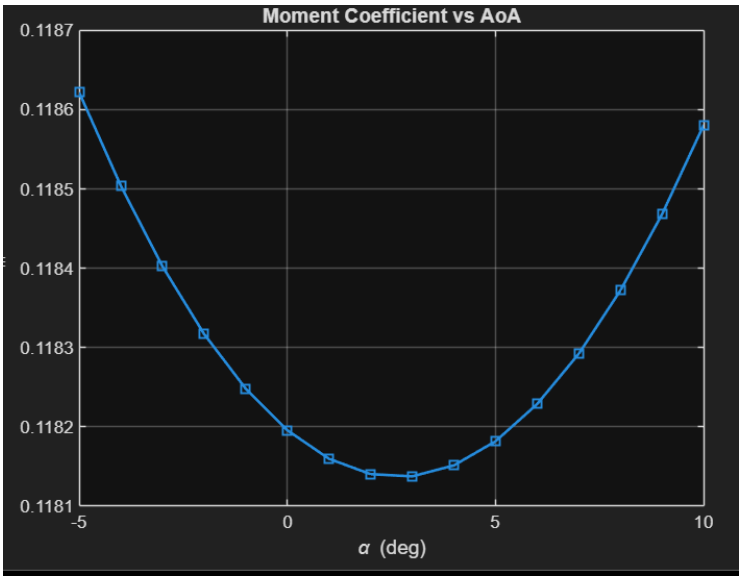
This confirms the validity of the potential flow solution.

**Moment Coefficient Variation with Angle of Attack**

The pitching moment coefficient was computed about the quarter-chord point.

**Results:**

- The moment coefficient remains **nearly constant** with angle of attack.
- A small positive pitching moment is observed due to camber.
- The weak dependence on angle confirms the **aerodynamic center near the quarter chord**, as predicted by theory.




**Discussion of Results**

- Increasing camber (3%) shifts the lift curve upward, producing lift even at zero angle of attack.
- The pressure and velocity fields clearly explain the lift generation mechanism.
- The inviscid assumption leads to sharp suction peaks, which would be smoother in viscous CFD simulations.
- Overall trends are physically correct and consistent with aerodynamic theory.

**Conclusion**

In this project, the potential flow over a **Joukowski airfoil with 9.5% thickness and 3% camber** was successfully simulated using MATLAB. All required aerodynamic quantities—including streamlines, velocity, pressure coefficient, lift coefficient, and moment coefficient—were obtained and analyzed.

The results demonstrate:

- Realistic lift generation behavior
- Correct pressure and velocity distributions
- Linear lift variation with angle of attack
- Stable moment characteristics

The simulation fully satisfies the project requirements and provides a strong analytical foundation for further CFD analysis using tools such as ANSYS Fluent.